

EDUCATION

Northeastern University

Candidate for M.S. Electrical and Computer Engineering Power Electronics & Power Systems Sep 2025 - April 2027
B.S. Electrical and Computer Engineering (GPA: 3.845) Sep 2022 - April 2026

TESLA

May 2026 - Aug 2026

Automation and Controls Engineering Intern, Elgin Illinois

- Manufacturing Controls Development for Optimus Robot
 - Developed Siemens S7 PLC ladder logic for two 2-axis (X-Z) servo gantry pick-and-place systems, coordinating servo positioning, machine sequencing, sensor interlocks, and pneumatic actuator control
 - Commissioned 8 automated manufacturing stations across 2 lines by validating 400+ I/O points across sensors, actuators, and field devices prior to production startup
 - Diagnosed and resolved mechanical, electrical, and PLC control logic faults using electrical schematics, wiring diagrams, and I/O diagnostics, reducing equipment startup delays
- Safety System Controls Design
 - Designed the electrical circuit integrating a Keyence safety laser scanner with a Banner light tower, configuring scanner safety zones and alarm logic to activate visual and audible warnings upon personnel entry into restricted areas
 - Designed and built an electrical control panel, including schematic development, component selection, wiring, enclosure assembly, and coordinating mechanical mounting for deployment of the safety system

JOHNSON AND JOHNSON MEDTECH

April 2025 - Aug 2025

Electrical Engineering Co-op, Monarch Surgical Robot, Santa Clara California

- Hardware Testing and Design
 - Reverse engineered unfamiliar PCB test fixtures from external company in Altium
 - Benchtop testing and troubleshooting with oscilloscope to trace signals and determine functionality
 - Prepared for redesign of Power Distribution Unit PCB for Robotic System
 - Performed EOL component analysis on BOM and requirements and specifications analysis using Jama
 - Coordinated changes with firmware, controls, systems, and service teams
- Prototyping support
 - Programmed off the shelf voice module with arduino to interface with PDU with circuit design
 - Tested on robotic system PDU and demoed capabilities for key research stakeholders
 - Coiled wires to adjust inductance and coupling factor to simulate EM field generated during manufacturing testing
 - Used oscilloscope and function generator to mimic EM signals as well as differential op amp and driver

BOSE CORPORATION

July 2024 - Dec 2024

Electrical Engineering Co-op, Concept Prototyping, Framingham Massachusetts

- Designed a Differential Microphone PreAmplifier PCB to support differential microphones for R&D
 - Designed schematic, PCB layout, and component selection in KiCad for a differential microphone preamplifier
 - Coordinated board assembly and material inventory with JLCPCB
 - Benchtop testing, troubleshooting, and redesign using oscilloscope
 - Developed a test plan to validate boards using Audio Precision and MDAQ
 - Enabled R&D to explore differential microphones to reduce noise in products and improve quality
 - Demonstrated approximately 40% improvement in SNR over the existing design, resulting in the procurement of 100 PCBs for R&D evaluation and prototyping
- Embedded Prototyping
 - Programmed M5 microcontroller to interface with bluetooth & neopixels in C++
 - Configured NFC sensor for concept speaker using Raspberry Pi
 - Demoed concept to key research stakeholders

TECHNICAL SKILLS

Languages: C++, Python, MATLAB, SystemVerilog, RISC-V Assembly

Hardware: Siemens S7 PLCs, Ladder Logic, Altium, KiCad, LTspice, Oscilloscope, Function Generator, Power Supply, DMM, Soldering, Audio Precision

Embedded: Arduino, Raspberry Pi, FPGA, Intel Quartus Prime, Linux

Design: SolidWorks, Git, Autocad, Agile

ACADEMIC RESEARCH

Northeastern University, Textiles Lab Researcher

Nov 2024 - Present

- Knitting Robot Researcher with PhD Meghan Hoffman & (c)PhD Amanda Shayna Ahteck
- Used python scripts and mathematical modeling based in graph theory to explore machine knitting robot capabilities
- Directed study during fall 2025 investigating incorporation of electronics into custom knitted fabrics for wearables

Research Experience for Undergraduates, University of Maryland in Baltimore County, MD *June 2023 - Aug 2023*

Mobile Pervasive Lab for Smart Computing and Sensing, Professor Nirmalya Roy

- Collaborated on “Addressing statistical heterogeneity in federated learning for sea ship datasets” with PhDs
- Independently developed a method utilizing Python scripts and feature extraction to homogenize heterogeneous data, resulted in improved object detection accuracy and training time reductions

Northeastern University, RIVeR Lab

Jan 2023 - Sept 2024

- UPLIFT Research Scholar contributing to CALiBER, an autonomous robotic platform under Prof. Taskin Padir

RELEVANT COURSES

Graduate Courses: Power Electronics, Electric Drives, Mobile Robotics

Undergraduate Courses: Electronics, Digital Design, Computer Architecture, Embedded Design, Circuits & Signals, Electromagnetics, Linear Systems, Networks, Algorithms, Integrated Circuits, Random Phenomena

LEADERSHIP & TEACHING EXPERIENCE

IEEE Honor Society Eta Kappa Nu, President

April 2025 - Present

- Led group in communication with nationals, planning networking events, coordinating tutoring schedules
- Tutored students in all fundamental electrical and computer engineering courses

Northeastern University Cornerstone of Engineering, Teaching Assistant

Sept 2023 - Present

- Graded and designed projects and exams to assess technical skills: C++, Arduino, AutoCAD, SolidWorks, MatLab

Northeastern University, Resident Assistant

Aug 2024 - May 2025

- Collaborated with a team to plan programs to create community among residents and resolve conflicts

INTERESTS

- Hobbies: Swimming (HS Team Captain, Lifeguard, Swim Instructor), music, reading, time with family and dog, knitting